```
Import tkinter as tk
From tkinter import messagebox
Import json
Import random



# ---------- FILE ----------
Def load_data(file):
  Try:
    With open(file, “r”) as f:
      Return json.load(f)
  Except:
    Return {}



Def save_data(file, data):
  With open(file, “w”) as f:
    Json.dump(data, f, indent=4)



Def generate_id():
  Return “CMP” + str(random.randint(1000, 9999))



# ---------- MAIN ----------
Root = tk.Tk()
```

```
Root.title(“Complaint System”)
Root.geometry(“450x550”)
Root.configure(bg=”#E8F6FF”)

Label_bg = “#E8F6FF”



# ---------- STYLES ----------
Btn_style = {
   “font”: (“Arial”, 11, “bold”),
   “bg”: “#27AE60”,
   “fg”: “white”,
   “width”: 22,
   “bd”: 0,
   “pady”: 6
}



Def clear_screen():
   For widget in root.winfo_children():
      Widget.destroy()



# ---------- HOME ----------
Def home():
   Clear_screen()
```

```
   Tk.Label(root, text=”Complaint System”,
         Font=(“Arial”, 18, “bold”),
         Bg=label_bg).pack(pady=30)

   Tk.Button(root, text=”User”, command=user_menu,
         **btn_style).pack(pady=10)
   Tk.Button(root, text=”Admin”, command=admin_panel,
         **btn_style).pack(pady=10)



# ---------- USER MENU ----------
Def user_menu():
   Clear_screen()

   Tk.Label(root, text=”User Panel”,
         Font=(“Arial”, 16, “bold”),
         Bg=label_bg).pack(pady=20)

   Tk.Button(root, text=”New       Complaint”,
          Command=user_details,
          **btn_style).pack(pady=10)

   Tk.Button(root, text=”View       Complaint”,
          Command=view_complaint,
          **btn_style).pack(pady=10)
```

```
    Tk.Button(root, text=”Back”, command=home,
              Bg=”#C0392B”, fg=”white”,
              Width=15).pack(pady=20)


# ---------- USER DETAILS ----------
Def user_details():
    Clear_screen()

    Tk.Label(root, text=”Enter Details”,
             Font=(“Arial”, 14, “bold”),
             Bg=label_bg).pack(pady=15)

    Name = tk.Entry(root, width=25)
    Phone = tk.Entry(root, width=25)
    Address = tk.Entry(root, width=25)

    Tk.Label(root, text=”Name”, bg=label_bg).pack()
    Name.pack(pady=5)

    Tk.Label(root, text=”Phone”, bg=label_bg).pack()
    Phone.pack(pady=5)

    Tk.Label(root, text=”Address”, bg=label_bg).pack()
    Address.pack(pady=5)
```

```
    Def next_step():
        User_complaint(name.get(), phone.get(), address.get())

    Tk.Button(root, text=”Submit”, command=next_step,
              **btn_style).pack(pady=15)

    Tk.Button(root, text=”Back”, command=user_menu,
              Bg=”#C0392B”, fg=”white”).pack()



# ---------- COMPLAINT ----------
Def user_complaint(name, phone, address):
    Clear_screen()

    Tk.Label(root, text=”Complaint Details”,
             Font=(“Arial”, 14, “bold”),
             Bg=label_bg).pack(pady=15)

    Ctype = tk.Entry(root, width=25)

    # BIG DESCRIPTION BOX
    Desc = tk.Text(root, height=5, width=30)

    Tk.Label(root, text=”Complaint Type”, bg=label_bg).pack()
    Ctype.pack(pady=5)
```

```
Tk.Label(root, text=”Description”, bg=label_bg).pack()
Desc.pack(pady=10)

Def register():
   Data = load_data(“complaints.json”)
   cid = generate_id()

   data[cid] = {
      “name”: name,
      “phone”: phone,
      “address”: address,
      “type”: ctype.get(),
      “description”: desc.get(“1.0”, tk.END).strip(),
      “status”: “Pending”,
      “estimated”: “”
   }

   Save_data(“complaints.json”, data)

   Messagebox.showinfo(“Success”, f”Your Complaint ID is {cid}”)
   Home()

Tk.Button(root, text=”Register       Complaint”,
        Command=register,
        **btn_style).pack(pady=15)
```

```
   Tk.Button(root, text=”Back”, command=user_menu,
          Bg=”#C0392B”, fg=”white”).pack()


# ---------- VIEW ----------
Def view_complaint():
   Clear_screen()

   Tk.Label(root, text=”Check Complaint”,
         Font=(“Arial”, 14, “bold”),
         Bg=label_bg).pack(pady=20)

   cid = tk.Entry(root, width=25)

   tk.Label(root, text=”Enter Complaint ID”, bg=label_bg).pack()
   cid.pack(pady=10)

   def check():
      data = load_data(“complaints.json”)
      resolved = load_data(“resolved.json”)

      c_id = cid.get()

      if c_id in data:
         c = data[c_id]
```

```
        messagebox.showinfo(
            “Status”,
            F”Status: {c[‘status’]}\nEstimated Time: {c[‘estimated’]}”
        )
    Elif c_id in resolved:
        Messagebox.showinfo(“Status”, “Complaint Resolved      “)
    Else:
        Messagebox.showerror(“Error”, “Invalid Complaint ID”)

  Tk.Button(root, text=”Check”, command=check,
          **btn_style).pack(pady=15)

  Tk.Button(root, text=”Back”, command=user_menu,
          Bg=”#C0392B”, fg=”white”).pack()


# ---------- ADMIN ----------
Def admin_panel():
  Clear_screen()

  Data = load_data(“complaints.json”)

  Tk.Label(root, text=”Admin Panel”,
          Font=(“Arial”, 16, “bold”),
          Bg=label_bg).pack(pady=15)
```

```
# FIX: show message if empty
If not data:
   Tk.Label(root, text=”No complaints available”,
         Bg=label_bg, fg=”red”).pack(pady=10)
Else:
   For cid, d in data.items():
      Tk.Label(root, text=f”{cid} – {d[‘type’]}”,
            Bg=label_bg).pack()

cid_entry = tk.Entry(root, width=25)
time_entry = tk.Entry(root, width=25)

tk.Label(root, text=”Complaint ID”, bg=label_bg).pack()
cid_entry.pack(pady=5)

tk.Label(root, text=”Estimated Days”, bg=label_bg).pack()
time_entry.pack(pady=5)

def set_time():
   cid = cid_entry.get()
   if cid in data:
      data[cid][“estimated”] = time_entry.get()
      save_data(“complaints.json”, data)
      messagebox.showinfo(“Done”, “Time updated”)

def resolve():
```

```
    cid = cid_entry.get()
    if cid in data:
        data[cid][“status”] = “Resolved”

        resolved = load_data(“resolved.json”)
        resolved[cid] = data[cid]

        del data[cid]

        save_data(“complaints.json”, data)
        save_data(“resolved.json”, resolved)

        messagebox.showinfo(“Done”, “Complaint Resolved”)

tk.Button(root, text=”Set        Time”,
          command=set_time,
          **btn_style).pack(pady=5)

Tk.Button(root, text=”Mark Resolved”, command=resolve,
          Bg=”#2980B9”, fg=”white”).pack(pady=5)

Tk.Button(root, text=”Back”, command=home,
          Bg=”#C0392B”, fg=”white”).pack(pady=10)


# ---------- START ----------
```

```
Home()

Root.mainloop()
```